MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10547240
(TARGET COUNTER, MEMORY AND READOUT - A4)

1. SCOPE

1.1 This specification covers a multi-layer printed circuit board, a subassembly of the Counter, Time Interval: 10547200 which is a modular component of the Range Finder, LASER, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Printed Circuit Assemblies
MIL-P-46843	Printed Circuit Assemblies, Design and Production of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10547240	Circuit Card Assembly - Target Counter, Memory and Readout - A4

PACKAGING DATA SHEET

10547240	Packaging of Target Counter Memory and Readout - A4

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The Circuit Card Assembly - Target Counter, Memory and Readout - A4, hereinafter referred to as assembly, shall be manufactured in accordance with Drawing F10547240 and drawings pertaining thereto.

3.3 General specification.- The assembly shall meet the following requirements, where applicable, of MIL-F-13926 and MIL-P-46843:

a. Dimensions and tolerances
b. Inorganic protection surface finisher
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926 and MIL-P-46843, the assembly shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.11.4 inclusive.

3.3.1.1 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface.

3.3.1.2 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that the proper identification markings on each part are visible.

3.3.1.3 Transistors.- Transistors are to be mounted perpendicular to the circuit board with spacers between the transistors and the circuit board.

3.3.1.4 Bend radius.- The minimum bend radii for formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.1.5 Component leads.- The minimum lead length for radial lead components shall be 4 times the lead diameter. The portion of the lead between a bend and the component body shall emerge from the body in a

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straight line radially from the component body and enter the plated hole approximately perpendicular to the board. Unless otherwise specified in the applicable drawing, leads of transistors mounted with a spacer directly into the printed circuit board, shall have a .062 inch of lead from the transistor base to any soldered joint. Component leads shall be preformed to suit mounting holes and clinched parallel to board after being inserted through the holes. There shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than .060 inch, including solder, beyond the side of the circuit board opposite the component. The leads shall not extend more than two lead diameters beyond the land area.

3.3.1.6 Soldering.- All solder joints shall conform to MIL-F-13926 and MIL-P-46843. There shall be a solder fillet between the board and the lead on the side of the board opposite the component. The fillet shall not obscure the contour of the clinched lead. A solder fillet is acceptable, but not required, on the component side of the board.

3.3.1.7 Plated-through connections.- All plated-through connections shall be filled with solder in order to eliminate dependence on plating itself for circuit continuity.

3.3.1.8 Electrical clearance.- A minimum of .010 inch shall be maintained between uncommon component leads, transistor cases and tabs.

3.3.1.9 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 pounds per square inch (psi). The temperature of the cleaning solution shall not exceed 125 degrees Fahrenheit (°F). After cleaning there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.1.10 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than the room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns.
b. Permissible moisture content limited to 2.12 grams per cubic foot (dew point 32°F maximum)
c. Dust count shall not exceed 1,000 particles per cubic foot

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3.3.1.11 Coating compound.

3.3.1.11.1 Masking.- Prior to coating, the areas specified on Drawing F10547240 shall be masked. Subsequent to coating, with masking removed, all specified areas shall be clean and free of coating.

3.3.1.11.2 Coating.- The assembly shall be coated with Insulating Compound, Electrical, Type PUR Polyurethane Resin conforming to MIL-I-46058.

3.3.1.11.3 Thickness.- Multiple applications shall be used with proper curing between each coat, until a minimum of .0010 inch thick to .0060 maximum thickness is achieved.

3.3.1.11.4 Cured compound.- The cured compound shall present a smooth continuous surface and shall pass from the mounted components to the circuit board with no break at their junctions. The cured compound shall meet the following requirements:

a. Compound shall be free of visible dust particles or other foreign matter.
b. It shall be smooth and uniform and have no globules or bare spots.
c. It shall be substantially free of bubbles.
d. There shall be no blisters or cracks.
e. It shall be transparent and shall not affect marking legibility.
f. It shall be capable of being removed in local areas for replacement of components by means of soldering iron or similar method.
g. Applied to a test board a 1/16 inch thickness, the compound shall not rupture or separate from the board when the board is bent into an arc of twelve inches radius.

3.4.1 Vibration.- The assembly shall withstand vibrations for 15 minutes in each of 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of the applied vibrations shall be .036 inch at lower frequencies and shall be maintained at, but not exceed, a 5G level at higher frequencies. Subsequent to vibration, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.5 Shock.- The assembly shall withstand a total of 6 shock impulses, one in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 millisecond. The peak amplitude of each shock impulse shall be 75Gs. Subsequent to shock, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

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3.6 Storage temperature.- The assembly shall withstand thermal stabilization at temperatures of -80°F and +160°F. Subsequent to exposure to the temperatures specified herein, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.7 Performance.

3.7.1 Operating temperature.- The assembly shall meet the requirements of 3.7.2 through 3.8 when thermally stabilized, at all temperatures from -65°F to +155°F inclusive.

3.7.2 Operation.- The functions of this assembly are as shown in Figure 1. With the following applied voltages

Pin K	+3.50 ± 0.2 Vdc
Pin G	+4.0 ± 0.2 Vdc (MEM)
Pin o	Power ground

this assembly shall operate as delineated below.

3.7.2.1 Start-stop flip-flop.

3.7.2.1.1 Reset input.- Application of a positive 2V ± 0.2V pulse of duration 1 microsec to pin B shall cause pin T to assume a DC ground potential (+0.1 ± 0.1V) and pin N to assume a DC potential of +4V ± 0.2V. Removal of 3.5V from pin k shall not change the DC level at pin N.

3.7.2.1.2 Start input.- A pulse of +1.7 ± 0.1V amplitude and 1.0 ± 0.5 microsec width applied to pin b shall effect the following results:

a. Pin j shall take on a ground potential (0.1 ± 0.1V) for a period of 135 ± 5 microsec and then return to a DC level of +3.1 ± 0.1V.

b. Pin T shall rise to a DC level of +1.5 ± 0.1V from its initial state of +0.1 ± 0.1V DC.

c. Pin N shall fall from its initial DC level of +4V ± 0.2V to a ground level of +0.1 ± 0.1V DC.

3.7.2.1.3 Overflow reset input.- Application of a positive 2.0V ± 0.2V pulse of duration 10 microsec to pin V shall cause pins T and N to revert to their initial states of 0.1 ± 0.1V and +4V ± 0.2 respectively.

3.7.2.1.4 Test start input.- Application of a positive 1.1V ± 0.1V pulse of 2.0 ± 0.5 microsec duration to pin R shall cause pin T to rise to a DC level of +1.5 ± 0.1V from its initial state of +0.1V ± 0.1V and pin N to fall from its initial DC level of +4V ± 0.2V to ground level of +0.1 ± 0.1V DC.

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3.7.2.2 Range mono. disable input.- Application of a DC level of +1.2V ± 0.1V to pin K shall cause pin j to take on a DC ground level of +0.1 ± 0.1V.

3.7.2.3 Stop pulse detection.- Application of a DC level of +2.5V ± 0.2V at pin O together with the application of a DC level of +1.2V ± 0.1V to pin X shall cause pin T to assume a DC ground (+0.1 ± 0.1V) potential and pin N to assume a DC level of +4.0 ± 0.2V when a 100 nanosec negative pulse falling from +5VDC to ground is applied to pin c. When this pulse is applied to pin c, a positive pulse of similar duration (100 nanosec) and amplitude 1.1 ± 0.1V shall be seen at pin M. With pin O at ground potential, (0.1 ± 0.1V) pins T, N and M shall not respond to pulses applied to pin c.

3.7.2.4 Pulse counting and memory.- This assembly shall have the capability to count the number of pulses falling from a 5 Vdc level to ground with duration of 100 nanoseconds applied to pin c after a positive pulse of 1.o±0.1 microseconds duration and 2.0±0.2V amplitude has been applied to pin B. The count of the pulses shall be decoded and provided to the Neon Tube Driver output pins S, I, H, F, J and d. With pin K maintained at a DC level of 1.2 ± 0.1V, the assembly shall be capable of counting the first 3 pulses applied to pin c. Subsequently, maintaing pin K at a DC level of 0.1 ± 0.1V shall allow the fourth and fifth pulses to be counted. With pin E simultaneously maintained at a DC level of 0.1 ± 0.1V, only the first 5 pulses applied to pin c shall be counted. With pin E maintained at a DC level of 1.2 ± 0.1V, the sixth pulse applied to pin c shall be counted. As the first pulse applied to pin c is counted, pin D shall assume a DC potential of 1.2V ± 0.2V. As the second pulse applied to pin c is counted, pin C shall assume a DC potential of 1.2V ± 0.2V. As the third pulse applied to pin c is counted, pin L shall assume a DC potential of 1.2±0.2V and a negative pulse of 1.4V±0.2V amplitude, duration 30 microsec, shall be generated at pin i. As the fourth pulse applied to pin c is counted, pin A shall assume a DC potential of 4.0V ± 0.2V. When pin E is maintained at a DC potential of +2V ± 0.2V and a fifth and sixth pulse are applied to pin c, a positive pulse 1.2V ± 0.2V amplitude, duration 30 microsec, shall appear at pin i when the sixth pulse is counted. When the +3.5V DC is removed from pin k, the outputs at pins N and j and the neon indicators S, I, H, F, J and d driver outputs shall maintain the voltage levels acquired prior to the removal of the +3.5Vdc.

3.7.2.5 Numerical indicator driver outputs.- When a positive DC level 1.4 ± 0.2V is applied to pin Y, W, Q or P, a path to system ground shall be established at pins Z, a, h, or U respectively.

3.7.2.6 Resistance measurement.- A 3K ohm resistance shall be measured from pin e to ground.

3.8 Electrical continuity.- The electrical wiring of the assembly shall conform to the electrical schematic drawing.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0% Defective			
101.	Electronic workmanship	3.3.1	Visual
102.	Spacing of parts	3.3.1.1	SME, Visual
103.	Symbols and markings	3.3.1.2	Visual
104.	Transistors	3.3.1.3	Visual
105.	Bend radius	3.3.1.4	SME, Visual
106.	Component leads	3.3.1.5	SME, Visual
107.	Soldering	3.3.1.6	Visual
108.	Plated-through connections	3.3.1.7	SME, Visual
109.	Electrical clearance	3.3.1.8	Visual
110.	Masking	3.3.1.11.1	Visual
111.	Cured compound	3.3.1.11.4	SME, Visual
MINOR: NONE			

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

TABLE II - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
112.	Reset input	3.7.2.1.1, 3.7.2	4.6.5.1
113.	Start input	3.7.2.1.2, 3.7.2	4.6.5.2
114.	Overflow reset input	3.7.2.1.3, 3.7.2	4.6.5.3
115.	Test start input	3.7.2.1.4, 3.7.2	4.6.5.4
116.	Range mono. disable input	3.7.2.2, 3.7.2	4.6.5.5
117.	Stop pulse detection	3.7.2.3, 3.7.2	4.6.5.6
118.	Pulse couting and mem.	3.7.2.4, 3.7.2	4.6.5.7
119.	Nixie driver outputs	3.7.2.5, 3.7.2	4.6.5.8
120.	Resistance measurement	3.7.2.6	4.6.5.9
121.	Electrical continuity	3.8	4.6.6

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NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.9
Drying	3.3.1.10
Coating, insulating compound	3.3.1.11.2

4.4 Special sampling.

4.4.1 General.- Three assemblies shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs & specs - visual
303.	General specification	3.3	MIL-F-13926 - visual, MIL-P-46843

NOTE: The tests in Table III shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- Three assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table II.

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TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration (+70°F and +80°F)	3.4	4.6.3
305.	Shock (+70°F and +80°F)	3.5	4.6.2
306.	Storage temperatures (-80°F and +160°F)	3.6	4.6.1.1
307.	Operating temperature (-65°F and +155°F)	3.7.2 through 3.8	4.6.4 through 4.6.5.8

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

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NOMENCLATURE	DESCRIPTION
1. D.C. Power Supply (2 req'd)	Range of 0-10V. current rating 500 ma.
2. D.C. Power Supply (1 req'd)	160 + 5V current rating 10 ma.
3. Oscilloscope	Risetime 10 nanoseconds or less, input impedance of 10 megohms or greater, sweep time of 100 nanoseconds per centimeter, sensitivity of .05 volts per centimeter.
4. Pulse generator (2 req'd)	Sufficient capability to generate a single positive or negative 2 volt pulse (ground reference) which can be manually controlled. The pulse width capability shall be 50 nanoseconds with rise and fall times not to exceed 25 nanoseconds. Output impedance of generator shall be 50 ohms. Generator must have manual trigger input and external trigger input with delay of 80 microseconds.

4.6 Test methods and procedures.

4.6.1 Environmental.- The assemblies shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the assemblies shall be subjected to the tests specified in 4.6.4 through 4.6.6.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.6.

4.6.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.5.

4.6.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The assembly shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.4. The vibration test shall then be performed in accordance with Method 514, Procedure I, Part 1, of MIL-STD-810.

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4.6.4 Operational temperatures.- The test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.7.1, utilizing the testing equipment indicated in 4.6.1.1. The assembly shall be exposed to each specified temperature for a period of 2 hours. At the end of each 2 hour period, and while still at the specified temperature, the assembly shall be subjected to the tests in 4.6.5 through 4.6.5.8. The assembly shall conform to the requirements of 3.7.1.

4.6.5 Operation procedure.- Connect the test circuitry as shown in Figure 2.

4.6.5.1 Reset input.- Apply a positive 2V ± 0.2V pulse of 1 microsec duration to pin B. Pin T shall assume a DC ground potential (+0.1 ± 0.1V) and pin N shall assume a DC potential of +4V ± 0.2V. Remove the 3.5V input from pin k. The voltage levels established above shall be unchanged. Reapply the 3.5V to pin k.

4.6.5.2 Start input.- Apply a positive 1.7 ± 0.1V pulse of 1 ± 0.5 microsec duration to pin b. Pin j shall take on a ground potential (+0.1 ± 0.1V) for a period of 135 ± 5 microsec and then return to a DC level of +3.1 ± 0.1V. Pin T shall rise to a DC level of +1.5 ± 0.1V. Pin N shall fall to a DC ground potential (0.1 ± 0.1V).

4.6.5.3 Overflow reset input.- Apply a positive 2.0 ± 0.2V pulse of 10 ± 1 microsec duration to pin V. Pin T shall assume a DC potential of +0.1 ± 0.1V and Pin N shall assume a +4.0 ± 0.2V potential.

4.6.5.4 Test start input.- Apply a positive 1.1V ± 0.1V pulse of 2.0 ± 0.5 microsec duration to pin R. Pin T shall assume a DC level of +1.5 ± 0.1V and Pin N shall assume a DC ground level (+0.1 ± 0.1V).

4.6.5.5 Range mono disable input.- Apply a DC level of +1.2 ± 0.1V to pin X. Pin j shall take on a DC ground level (+0.1 ± 0.1V).

4.6.5.6 Stop pulse detection.- Apply the +1.2 ± 0.1V DC level to pin X and a DC level of +2.5 ± 0.2V to pin O. Apply a 100 nsec negative pulse falling from 5V to ground to pin c. A positive pulse 100 nsec wide and 1.1 ± 0.1V in amplitude shall appear at pin M. Pin T shall be at a DC ground level (+0.1 ± 0.1V) and pin N shall have a DC potential of +4V ± 0.2V. Remove the positive potential from pin O and connect pin O to ground. Apply a pulse of 1.7 ± 0.1V amplitude and 1 ± 0.1 microsec duration to pin b. Pin T goes to 1.5V ± 0.2V and N goes to 0.1 ± 0.1V. Apply a 100 ns negative pulse to pin c. Pin M shall maintain a DC level of +0.5 ± 0.2V and there shall be no pulses appearing at pin M. Pin T shall maintain a DC level of +1.5 ± 0.1V and pin N shall maintain a DC ground level (+0.1 ± 0.1V).

4.6.5.7 Pulse counting and memory.- Apply a positive 2V ± 0.2V pulse to pin B. No neon indicator lamps shall be lighted. Remove 3.5V from pin k. The neon indicator lamps shall remain unlighted. Reapply 3.5V to k. Apply a DC level of 1.2 ± 0.1V to Pin X. This level shall be maintained for the remainder of this test. Apply 1 negative pulse as described above to pin c. The neon indicator connected between pins I and H only shall be lighted, and pin D shall have a DC level of 1.2V ± 0.2. Remove 3.5V from pin k. This indicator lamp shall remain lighted. Reapply the 3.5V to pin k. Apply a positive 2V pulse

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to pin B. Apply 2 pulses to pin c. The neon indicator connected between pins H and F only shall be lighted and pin C shall have a DC level of 1.2V ± 0.2V. Remove the 3.5V from pin k. This indicator only shall remain lighted. Reapply the 3.5V to pin k and apply a positive 2V pulse to pin B. Apply 3 negative pulses to pin c. As the third pulse is applied, a negative pulse 1.4V ± 0.2V in amplitude and 30 microsec duration shall be observed at pin i. The neon indicator connected between pins F and J only shall be lighted and pin L shall have a DC level of 1.2V ± 0.2V. Remove the 3.5V from pin k. This indicator only shall remain lighted. Reapply the 3.5V to pin k and apply a positive 2V pulse to pin B. Apply 4 negative pulses to pin c. The neon indicator connected between pins J and d only shall be lighted and pin A shall have a DC level of 4V ± 0.2V. Remove the 3.5V from pin k. This neon indicator only shall remain lighted. Reapply the 3.5V to pin k, and apply a positive 2V pulse to pin B. Apply 5 negative pulses to pin C. The neon indicator connected between pins d and S only shall be lighted. The application of additional pulses to pin c shall cause no change in the neon indicator lamps. Remove the 3.5V from pin k. Only this neon indicator lamp shall remain lighted. Reapply the 3.5V to pin k. Remove the ground connection from pin E and apply 2V to pin E. As negative pulses are applied to pin c, the neon indicators will light and cycle in the sequence 0 through 5. As the fifth target light is extinguished, a positive pulse 1.2V ± 0.2V in amplitude and 30 microsec in duration shall be observed at pin i.

4.6.5.8 Numerical indicator driver outputs.- Apply +3.5V through 3K ohm resistor to pin Y. The "0" shall light on the nixie tube. Apply +3.5 through a 3K ohm resistor to pin W. The "1" shall light on the neon tube. Apply +3.5V through a 3K ohm resistor to pin Q. The "5" shall light on the nixie tube. Apply +3.5V through a 3K ohm resistor to pin P. The "7" shall light on the neon tube.

4.6.5.9 Resistance measurement.- Measure 3K ohms to ground from pin e.

4.6.6 Electrical continuity.- Continuity shall be inspected by standard measuring equipment (SME) and visually examined for compliance with Drawing F10547240 and as specified in 3.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10547240. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Circuit Card Assembly, Target Counter, Memory and Readout - A4 generates the signal to start and stop the counter during ranging. Additional circuitry is used to determine the number of targets received. It is a subassembly of the Counter, Time Interval, which is a module of the Range Finder, LASER, AN/GVS-3.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Applicable packaging data sheet
c. Level of protection required
d. Provisions for First Article testing

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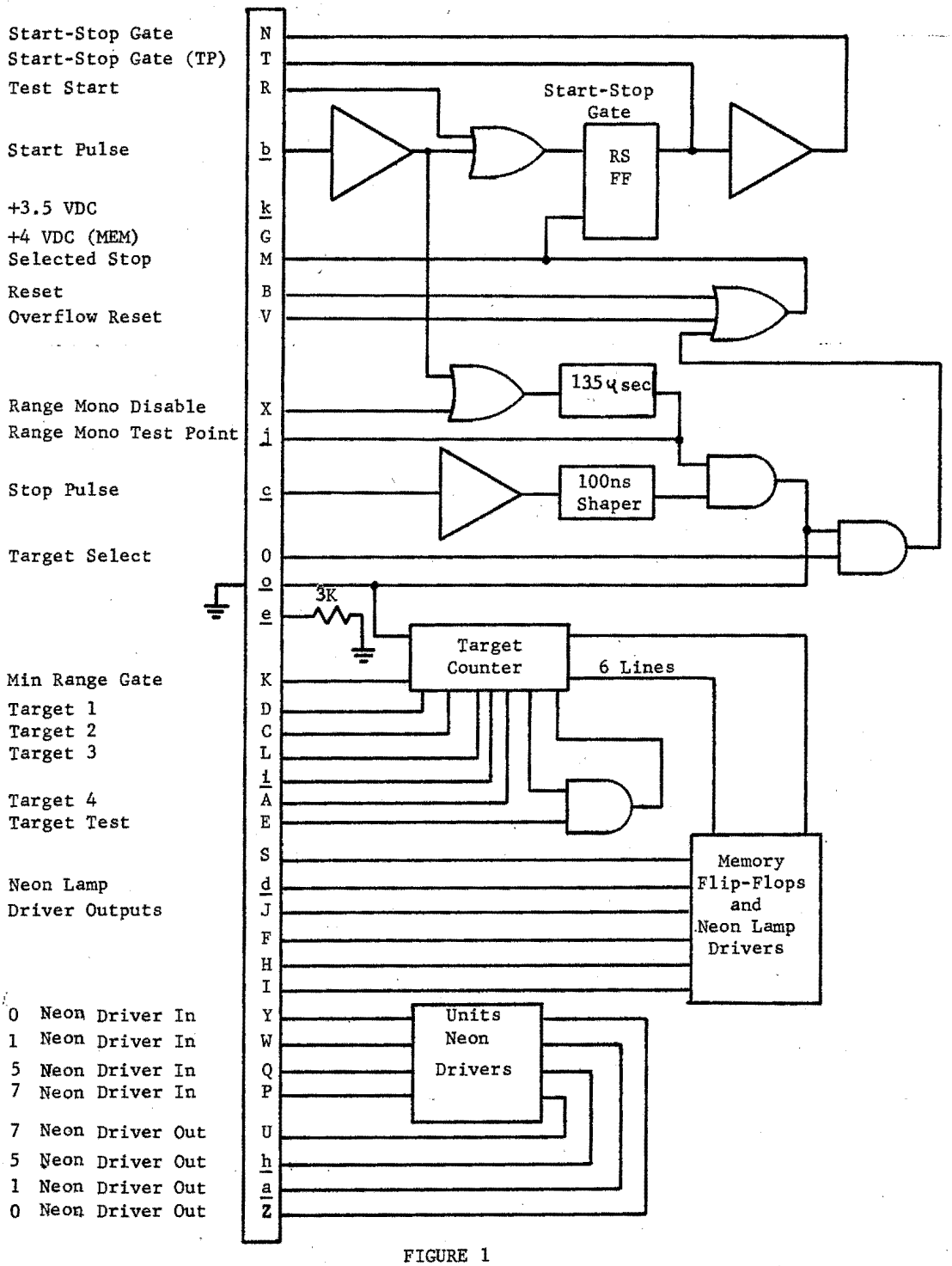
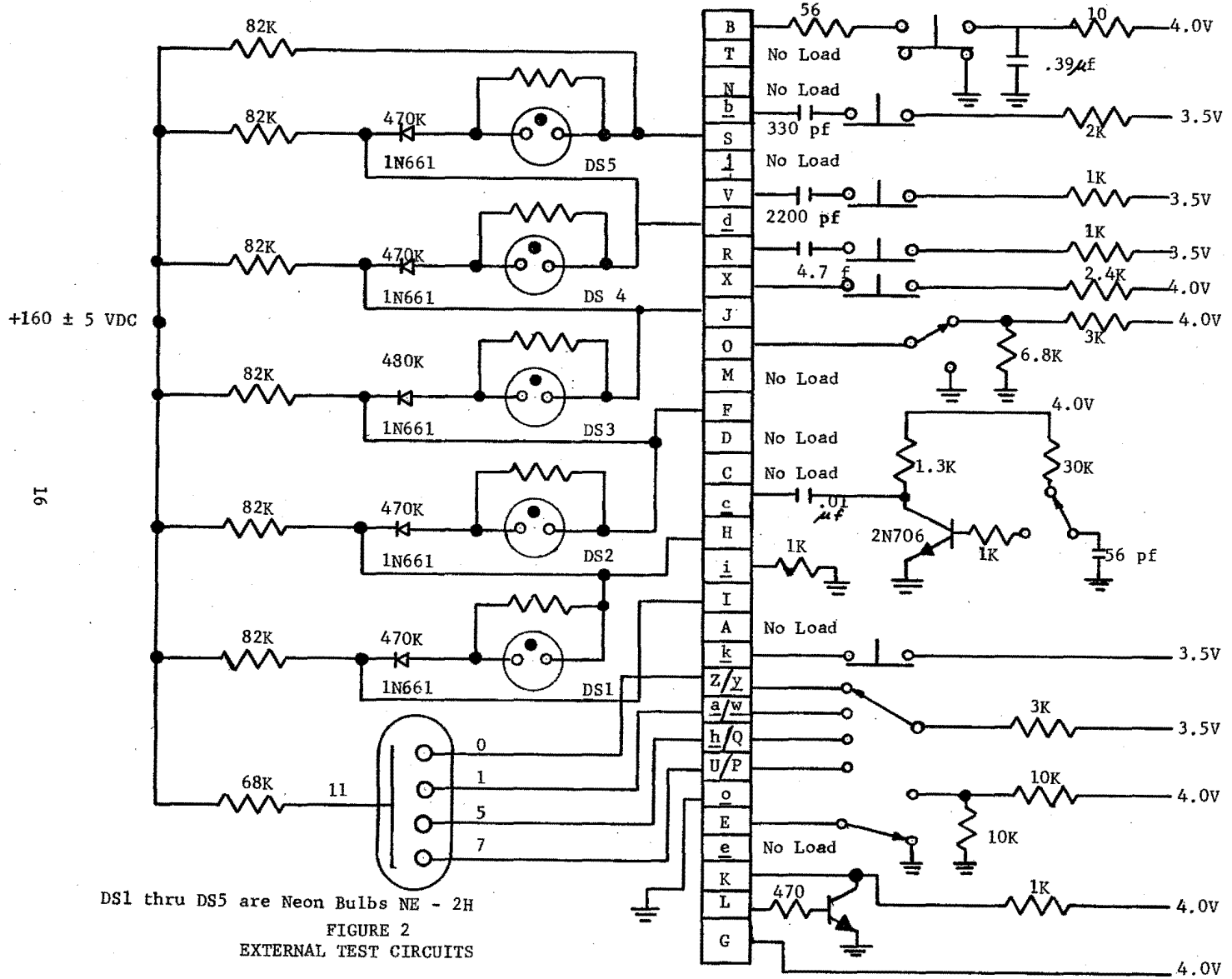


FIGURE 1

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DS1 thru DS5 are Neon Bulbs NE - 2H

FIGURE 2
EXTERNAL TEST CIRCUITS

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